StyleIQ: Dialog-Guided Vision-Based Outfit Recommendation System with Body-Aware Visual Pop-Up Overlays

Introduction

Recent advancements in computer vision and human computer interaction have enabled systems to analyze visual content and respond intelligently to user input. These technologies have been widely applied in domains such as surveillance, healthcare, retail, and digital media. In the fashion industry, computer vision has become increasingly relevant due to its potential to automate clothing analysis, enhance personalization, and improve user decision-making in outfit coordination.

Despite the availability of numerous fashion applications and online styling platforms, many users still struggle to coordinate outfits that are visually harmonious. This difficulty often stems from subjective fashion judgment, limited styling knowledge, and the absence of tools that analyze what the user is wearing. Most existing fashion recommendation systems rely heavily on browsing large catalogs or manually selecting preferences, rather than intelligently interpreting clothing from images.

StyleIQ is proposed as a **dialog-guided, vision-based fashion recommendation system** that assists users in coordinating outfits by analyzing visual input and generating body-aware recommendations. The system allows users to type a description or instruction in a dialog box specifically emphasizing a clothing item they want to focus on (e.g., *"emphasize blue skirt"*). This emphasized item becomes the foundation for all outfit recommendations.

StyleIQ integrates **image-based clothing analysis**, **rule-based mix-and-match logic**, and **AR-style visual pop-up overlays** displayed on a live webcam feed or uploaded image. Rather than aiming for perfect augmented reality fitting, the system prioritizes **clear body-based placement and recommendation logic**, ensuring feasibility on standard laptops while still demonstrating innovation.

Problem Statement

Coordinating outfits is a common challenge for many individuals, particularly when attempting to match colors, styles, and accessories in a visually pleasing way. This challenge is intensified by time constraints, lack of fashion confidence, and uncertainty about compatibility between clothing items.

Current fashion recommendation applications often suffer from the following limitations:

- Heavy reliance on manual browsing and user-selected filters,
- Lack of intelligent analysis of the user's current outfit,
- Minimal user control over which clothing item should be emphasized,
- Limited visual feedback showing how recommendations relate to the user's body.

Additionally, while advanced augmented reality solutions exist, they typically require complex tracking algorithms, large datasets, and high-performance hardware. These requirements make such systems impractical for student-level development and everyday accessibility.

There is a need for a system that:

- Uses computer vision to analyze clothing visually,
- Allows users to guide recommendations through typed dialog input,
- Displays recommendations in a body-aware and intuitive manner,
- And remains realistic to implement within limited resources.

Objectives of the Study

The primary objective of this study is to design and develop **StyleIQ**, a dialog-guided, vision-based outfit recommendation system that provides visually grounded and user-centered styling assistance.

Specifically, the study aims to:

1. **Design a dialog box interface** that allows users to type and emphasize a specific clothing item as the anchor for outfit recommendations.
2. **Analyze outfit images or live webcam input** to extract dominant clothing colors and identify clothing categories.
3. **Implement an emphasize-one-item recommendation logic**, ensuring that all suggested items are coordinated around the user-selected clothing piece.
4. **Develop a controlled mix-and-match recommendation engine**, limited to five predefined items per clothing category to maintain feasibility and clarity.
5. **Overlay recommended clothing items as visual pop-ups** on a live webcam feed or uploaded image, positioned according to basic body regions.
6. **Provide hair and makeup recommendations as text-based styling instructions**, displayed below the visual output.
7. **Support both live camera-based interaction and photo-based outfit analysis** for flexible system usage.
8. **Evaluate the system's effectiveness** based on recommendation relevance, visual clarity, and user satisfaction.

Background and Related Literature

Computer vision research in fashion has demonstrated that visual attributes such as color, shape, and category can be effectively extracted from clothing images. Datasets such as **DeepFashion** provide annotated clothing images that support research in garment recognition, while tools like **OpenCV** are widely used for image preprocessing and color extraction.

Existing fashion recommendation systems typically fall into two categories: large-scale product recommendation platforms and trend-based styling applications. While effective in commercial environments, these systems often lack transparency and user control, making it difficult for users to understand why certain items are recommended.

Studies on rule-based fashion coordination suggest that even simple color harmony and category-matching rules can produce useful styling suggestions, especially in systems with limited datasets. However, many existing implementations do not integrate user-guided emphasis or body-aware visual presentation.

StyleIQ builds upon these concepts by combining dialog-based interaction, image analysis, and annotation-based visual overlays to create a system that is interpretable, user-guided, and feasible.

Methodology

System Overview

The proposed system follows a modular design consisting of:

- Dialog input module,
- Image acquisition module,
- Clothing analysis module,
- Recommendation engine,
- Visual overlay and user interface module.

Input Modes

StyleIQ supports two primary input modes:

1. **Live Webcam Mode**
   The user stands in front of a webcam, allowing the system to analyze their outfit and display recommendations relative to their body.
2. **Photo-Based Mode**
   The user uploads an image of their outfit for analysis. This mode allows styling without real-time camera interaction.

Dialog Box Interaction

At the beginning of each session, the system displays a **dialog box** where the user types a description or emphasis command, such as:

- *“emphasize blue skirt”*
- *“focus on red top”*

This input determines:

- Which clothing item is prioritized,
- Which category becomes the anchor,
- How the recommendation logic is applied.

Clothing Analysis

The system analyzes visual input to:

- Identify the emphasized clothing region,
- Extract dominant color information,
- Determine the clothing category (top, bottom, shoes, bag, accessories).

Color extraction is performed using basic image processing techniques, while category identification relies on predefined mappings and visual cues.

Recommendation Engine

The recommendation engine is **rule-based**, operating within a limited dataset.

- Five predefined women's clothing items are assigned to each category:
  - Tops
  - Bottoms
  - Shoes
  - Bags
  - Accessories

Recommendations are generated using:

- Color harmony rules,
- Category compatibility,
- Visual balance considerations.

All recommendations are restricted to the curated item pool, ensuring consistent and explainable results.

Visual Pop-Up Overlay

Recommended items are displayed as **annotation-based pop-up images** overlaid on the camera feed or uploaded image.

- Tops appear on the upper body region,
- Bottoms appear on the lower body,

- Shoes appear near the feet,
- Bags and accessories appear near the torso or upper body.

Precise alignment is not required; the goal is **logical body-based placement**.

Hair and Makeup Recommendations

Hair and makeup suggestions are displayed as **text-only instructions** below the visual output. This design choice avoids the complexity of facial overlays while still providing complete styling guidance.

Project Scope and Limitations

The system focuses on:

- Women's clothing only,
- One emphasized clothing item per session,
- Five predefined items per category,
- Basic visual analysis and rule-based logic.

The system does not support:

- Perfect AR fitting,
- Fabric texture analysis,
- Trend forecasting,
- Multi-person detection,
- Voice or gesture interaction.

Expected Outcomes

The study is expected to produce:

- A functional dialog-guided fashion recommendation system,
- Visually grounded outfit suggestions,
- Improved user confidence in outfit coordination,
- Positive feedback regarding usability and clarity.

Feasibility and Risk Assessment

The system is feasible on a standard laptop using open-source tools. Potential risks include lighting variability and imperfect body placement, which are mitigated through controlled assumptions and simplified design goals.

Conclusion

StyleIQ presents a practical and innovative approach to fashion recommendation by combining dialog-guided input, computer vision, and body-aware visual overlays. By prioritizing clarity, feasibility, and user control, the system serves as a strong foundation for future smart fashion technologies while remaining achievable for undergraduate implementation.